

DATA STRUCTURES - Examination 2019

Total Marks: 70

SECTION A

1.

Attempt all questions in brief.

- (a) Define array and linked list.
- (b) What is a queue? Give two applications.
- (c) State the time complexity of quick sort.
- (d) Define a doubly linked list.
- (e) What is a binary tree?
- (f) Differentiate between BFS and DFS.
- (g) What is a hash table?

SECTION B

2.

Attempt any three of the following:

- (a) Differentiate between array and linked list with suitable examples.
- (b) Explain circular queue with insertion and deletion algorithms.
- (c) Explain inorder, preorder and postorder tree traversal with examples.
- (d) Explain collision resolution in hashing.
- (e) Explain merge sort and analyze its complexity.

SECTION C

3.

Attempt any one part of the following:

- (a) Explain insertion and deletion in a binary search tree.
- (b) Explain BFS and DFS traversal of a graph with an example.

4.

Attempt any one part of the following:

- (a) Write the binary search algorithm and discuss its complexity.
- (b) Write a short note on B-trees.